

# DEVELOPMENT OF DAMAGE IN A 2D WOVEN C/SiC COMPOSITE UNDER MECHANICAL LOADING: I. MECHANICAL CHARACTERIZATION

Gérald Camus,<sup>a</sup> Laurent Guillaumat<sup>a</sup> & Stéphane Baste<sup>b</sup>

<sup>a</sup>Laboratoire des Composites Thermostructuraux, UMR-47 (CNRS-SEP-UBI), 3 Allée de La Boétie, 33600 Pessac, France

<sup>b</sup>Université de Bordeaux-I, Laboratoire de Mécanique Physique, URA-867, 351 Cours de la Libération, 33405 Talence, France

(Received 23 May 1995; revised 17 June 1996; accepted 27 June 1996)

## Abstract

*An investigation has been undertaken to determine the damage mechanisms and the associated mechanical response of a 2D reinforced composite of carbon fibers in an SiC CVI-processed matrix subjected to uniaxial tensile and compressive loadings at room temperature. Under tension loading, an extended non-linear stress/strain response was evidenced and related to a multi-stage development of damage involving transverse matrix microcracking, bundle/matrix and inter-bundle debonding as well as thermal residual stress release. This tensile behavior proved to be damageable-elastic with respect to a fictitious thermal-stress-free origin of the stress/strain axis lying in the compression domain. In compression, after an initial stage involving closure of the thermal microcracks present from processing, the composite displayed a linear-elastic behavior until failure. The extent of damage over the material was characterized quantitatively at the microscale by the decrease of the average transverse microcrack spacing and at the macroscale by the decrease of both the longitudinal Young's modulus and the in-plane Poisson's ratio.*

© 1997 Elsevier Science Limited

**Keywords:** ceramic-matrix composites, 2D C/SiC, mechanical behavior, damage mechanisms, moduli reduction

## 1 INTRODUCTION

Carbon-fiber/SiC-matrix (C/SiC) composites processed by chemical vapor infiltration (CVI) have been developed for high-temperature structural applications such as air-breathing engines, hot-gas valves and aerospace thermal structures.<sup>1,2</sup> These materials represent a good trade-off between the high-temperature capability of the carbon fibers and the high modulus/excellent oxidation resistance of the SiC

matrix which, when used in conjunction with additional surface seal-coatings, prevent oxidation of the carbon fibers.<sup>3,4</sup> Like most continuous-fiber-reinforced ceramic-matrix composites (CMCs), C/SiC composites exhibit a non-catastrophic failure coupled to an extended non-linear behavior under tension loading.<sup>2,5,6</sup> This behavior is the consequence of the onset of various energy-dissipating mechanisms such as matrix microcracking, fiber/matrix debonding and/or sliding and fiber pull-out.<sup>7,8</sup> Characterization of the non-linearity of the deformational response once the various energy-dissipating entities, collectively referred to as damage, have taken place provides useful input to structural design. This is particularly true in the case of C/SiC composites for which matrix microcracks are present from processing, because of the large coefficient of thermal expansion (CTE) mismatch that exists between the fibers and the matrix.<sup>6,9</sup>

Previous studies related to the room-temperature mechanical behavior of CVI-processed C/SiC composites have dealt primarily with fracture,<sup>10,11</sup> fatigue<sup>5,12</sup> and oxidation effects.<sup>6</sup> A detailed investigation of damage and the associated stress/strain non-linearity remains to be conducted. The objective of this two-part paper is thus to report an experimental study aimed at identifying through two complementary and compatible methods, mechanical and ultrasonic, the parameters fully characterizing anisotropic damage of a C/SiC composite. In this first part, results obtained from classical strain measurements (on the basis of the mechanical method applied to tensile and compressive tests), along with microstructural observations of damage-induced changes, are reported and discussed. The complete variations of each term of the stiffness tensor required to fully characterize anisotropic damage and obtained through the ultrasonic method, as well as comparisons between mechanical and ultrasonic measurements, are treated in Part II.<sup>13</sup>

## 2 EXPERIMENTAL

### 2.1 Materials

The composite material used in the present investigation was processed (by SEP, Société Européenne de Propulsion) by chemical vapor infiltration of SiC into a 2D fibrous preform consisting of a stack of ex-PAN carbon-fiber woven fabrics (with an average of 3000 fibers per bundle). Before infiltration by the matrix, the carbon fibers were coated with a pyrocarbon interphase of mean thickness  $\sim 1 \mu\text{m}$  in order to enhance the desired non-linear/non-catastrophic tensile behavior.<sup>14</sup> These processing steps resulted in a material having a density close to 2, a fiber content of approximately 40 vol.% and a residual porosity inherent to the CVI process in the range 10–15 vol.%. Typical properties of the SiC CVI matrix and the carbon fibers are listed in Table 1.

### 2.2 Test procedure

The mechanical characterization of the composites was conducted at room temperature under both tension and compression loading using flat dogbone-shaped specimens of 125 mm total length. The dimensions, geometry and orientation of the specimens are presented in Fig. 1. In order to maintain consistency with Part II, the usual reference axis orientation used in acoustic engineering has been retained (i.e. axis 3 lying parallel to the loading direction). Tests were performed on a servo-hydraulic machine equipped with pneumatic grips, at a constant slow strain rate of  $0.04\% \text{ min}^{-1}$ , i.e. under quasi-static conditions.  $0^\circ/90^\circ$  strain gauges were glued on the opposite sides of each specimen to monitor the axial and transverse strains. For each type of test, one specimen was tested monotonically to failure whereas two other specimens (only one in the case of the compression tests) were tested with interposed loading/unloading cycles in order to follow the occurrence and extension of damage over the material. For a similar purpose, acoustic emission (AE) was also recorded continuously in the cumulative count mode during each test, by using a

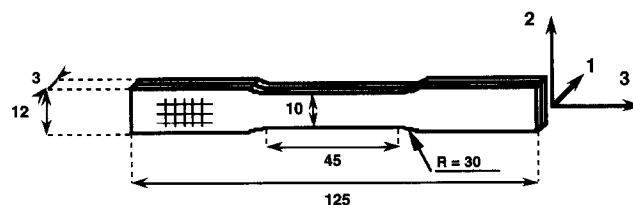


Fig. 1. Dimensions (in mm), geometry and orientation of specimens used for the tensile and compressive tests.

piezo-electric sensor in direct contact with the specimens.

In addition, one specimen was tested in tension/compression, i.e. each unloading step was prolonged in the compression domain down to  $-250 \text{ MPa}$ , to check the precise nature of the anelastic residual strains.

Changes in the microstructure induced by the onset and extension of damage were examined by optical microscopy: (i) *in situ* during the tests, on specimen edges and/or major faces that had been polished (with diamond paste to a  $1 \mu\text{m}$  finish) and (ii) on samples cut from specimens tested to failure and/or interrupted at two different load levels, 150 and 250 MPa, in order to follow the extension of damage in the bulk of the composite as well as in the plane (1,2) perpendicular to the loading direction. Average crack spacings were determined over lengths of the order of 15 mm from five different bundles for each specimen.

Finally, fracture surfaces of broken tensile specimens were examined with scanning electron microscopy (SEM).

## 3 RESULTS AND DISCUSSION

### 3.1 Material characterization

The observation by optical microscopy of polished samples cut from the as-received material reveals, in addition to the residual porosity already mentioned, an extended array of matrix microcracks (Fig. 2).

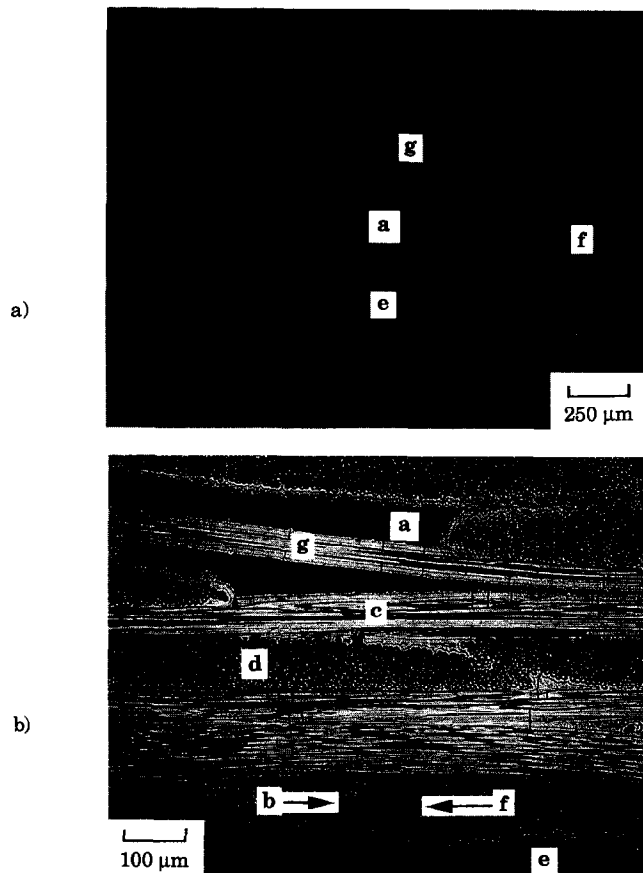
Two different populations may be observed regarding the residual porosity: (i) large macropores present in the SiC matrix between the fiber bundles

Table 1. Typical thermomechanical properties of the ex-PAN carbon fibers and SiC CVI matrix

| Constituent <sup>a</sup> | $\rho$<br>( $\times 10^3 \text{ kg m}^{-3}$ ) | $E$<br>(GPa) | $\nu$ | $\sigma^R$<br>(GPa) | $\epsilon^R$ (%) | $\alpha$<br>( $\times 10^{-6} \text{ K}^{-1}$ ) |
|--------------------------|-----------------------------------------------|--------------|-------|---------------------|------------------|-------------------------------------------------|
| C fiber                  | 1.8                                           | 230          | 0.3   | 3.5                 | 1.5              | Axial 0<br>Radial 10                            |
| SiC matrix               | 3.2                                           | 350          | 0.2   | —                   | 0.035            | 4.6                                             |

<sup>a</sup> Values given are estimated by SEP.

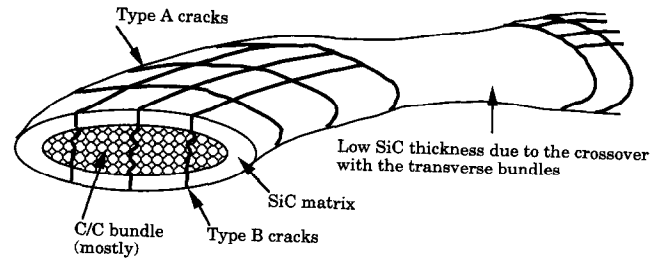
$\rho$ ,  $E$  and  $\nu$  are the density, Young's modulus and Poisson's ratio, respectively;  $\sigma^R$  and  $\epsilon^R$  are the stress and strain to rupture, respectively;  $\alpha$  is the coefficient of thermal expansion.



**Fig. 2.** Typical microstructure of the as-received 2D C/SiC composite revealing residual porosity and an array of matrix microcracks: (a) in the plane of the plies, plane (2,3); (b) perpendicular to the plies, plane (1,3). a, Macropores; b, micropores; c, longitudinal bundle; d, transverse bundle; e, SiC matrix; f, type B cracks; g, type A cracks.

and (ii) micropores present inside each fiber bundle. It should also be mentioned that these bundles appear to consist mainly of carbon fibers embedded in their carbon interphase with a very low matrix content, very likely as a consequence of both the relatively significant thickness of the carbon interphase and the small variation of the carbon fiber diameter.

Microcracks are related to the significant thermal expansion mismatch that exists between the fibers and the matrix (see Table 1). Thus, even though the carbon interphase may act as a buffer layer, the matrix still encountered both axial and radial tensile residual stresses upon cooling from the processing temperature, resulting in microcracking (and partial debonding, as previously established from transmission electron microscopy (TEM) observations<sup>9</sup>). Two distinct families of cracks, both exhibiting a bidirectional pattern reflecting the 2D texture of the reinforcement, may be distinguished in Fig. 2. To aid understanding, these two families of cracks are also schematically represented in Fig. 3 for a single direction of reinforcement, i.e. a fiber bundle with its



**Fig. 3.** Schematic representation of the thermal microcrack array existing in each individual bundle of the as-received 2D C/SiC composite.

surrounding SiC matrix fictitiously isolated from the material.

The first family of cracks, referred to as A, occurs in the matrix-rich areas present between the fiber bundles. A large portion of these cracks appears to be initiated at the sharp singularities of the macropores present between the fiber bundles and acting as stress concentrators. The cracks then tend to deviate perpendicular to the fiber bundles and are arrested at their periphery, most likely because of their low matrix content (Fig. 2(a)). Observations performed in the longitudinal plane (1,3) (i.e. through the thickness, Fig. 2(b)) reveal a significant variation in the spacing of these type A cracks depending on the local environment of the longitudinal bundles, i.e. isolated or in contact with two transverse bundles. In the former case, weakly opened microcracks revealed only by chemical etching are proved to be present between two larger cracks generated by macropore singularities, leading to an average spacing of the order of  $150\ \mu\text{m}$ . Conversely, in the latter case the bundle is locally free of cracks, so that the overall distribution of type A cracks along each bundle may be depicted by alternating spacings of  $150$  and  $600\ \mu\text{m}$ .

The second family of cracks, referred to as B, is found parallel to the fibers, inside each fiber bundle as well as through the surrounding SiC matrix (when present). These type B cracks appear to be homogeneously distributed with an average spacing of the order of  $140\ \mu\text{m}$  and are largely opened when located inside the bundles. It is not yet clear whether these cracks are initiated inside the bundles or in the surrounding matrix, or even possibly proceed from the propagation of type A cracks in the adjacent transverse bundles. Indeed, such discontinuities present inside the bundles, because of their large interphase content, may be regarded as being half way between microcracking and debonding.

### 3.2 Stress/strain response

#### 3.2.1 Tensile behavior

The room-temperature tensile properties of the C/SiC composite are summarized in Table 2, along with the

**Table 2. Measured tensile and compressive properties of a 2D C/SiC composite**

| Property                                              | Tension           | Compression       |
|-------------------------------------------------------|-------------------|-------------------|
| Young's modulus, $E_3$ (GPa)                          | $115 \pm 12$      | $140 \pm 5$       |
| Poisson's ratio, $\nu_{32}$                           | $0.07 \pm 0.005$  | $0.085 \pm 0.005$ |
| Stress to rupture, $\sigma^R$ (MPa)                   | $455 \pm 15$      | $520 \pm 10$      |
| Longitudinal strain to rupture, $\varepsilon_3^R$ (%) | $0.92 \pm 0.07$   | $0.4 \pm 0.03$    |
| Transverse strain to rupture, $\varepsilon_2^R$ (%)   | $0.025 \pm 0.008$ | 0.035             |

compressive properties which are addressed in Section 3.2.2. A typical stress/strain tensile curve including interposed loading/unloading cycles, the recording of the total acoustic emission count and the transverse strain response is plotted in Fig. 4.

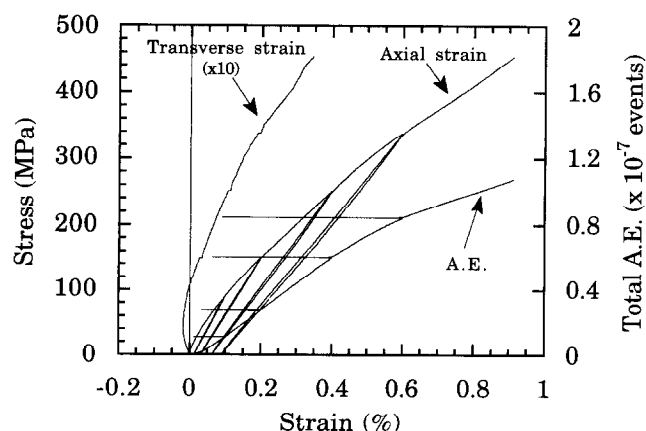
The composite behaves as a typical damageable material, exhibiting an extended non-linear stress/strain domain up to rupture. Moreover, as a consequence of the presence of the processing-related initial damage (hereafter referred to as thermal microcracks), non-linearity starts almost from the onset of loading. Nevertheless, an apparent initial Young's modulus of about 115 GPa could be obtained from the beginning of the stress/strain curve. As the AE signal actually starts to increase above an applied stress of 30–40 MPa, it is thus very likely that the mechanisms involved in this very first part of the stress/strain curve are partial re-opening of the thermal microcracks along with the global elastic response of the composite.

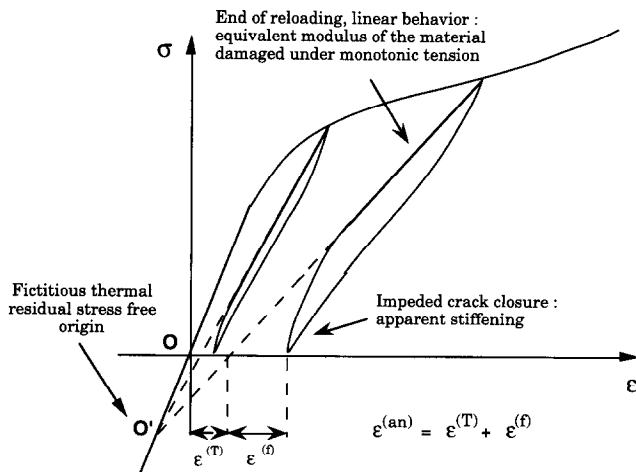
Three different domains corresponding to three different steps in the onset/extension of the damage phenomena (discussed later in Section 3.4) may then be noticed from both the  $\sigma/\varepsilon$  curve and the curve of

cumulative AE events versus strain. The first domain is characterized by a nearly constant increase in the accumulation of AE events and corresponds to multiplication of the matrix microcracking and associated partial debonding until saturation. A second domain involving crack opening and debonding propagation, characterized by a noticeable decrease in the rate of accumulation of AE events, then occurs. The last part of the curve displays a slight stiffening followed by a linear domain and corresponds to the elongation of the fibers, which are carrying the entire load. Rupture then occurs at an average stress of 455 MPa for a corresponding average strain of 0.92%; the associated moderate scatter is  $\pm 15$  MPa and  $\pm 0.07\%$ , respectively.

The recording of the in-plane transverse strain allowed us to measure an apparent initial Poisson's ratio of 0.07. A continuous increase of the transverse strain corresponding to a continuous decrease of the Poisson's ratio then occurs rapidly once damage takes place and propagates. Reversal in the transverse strain caused by the occurrence of matrix microcracking and fiber/matrix partial debonding has been commonly observed with many CMCs and was attributed to the progressive release of stresses (imposed and residual, if any) in the matrix at and near the crack surfaces.<sup>15,16,17</sup> The matrix thus sees its lateral Poisson's contraction being progressively reduced. In most CMCs, however, the transverse strain is commonly observed to reach a maximum value at matrix cracking saturation and then to remain constant until final rupture. Conversely, in the present case of the C/SiC composite, the transverse strain appears to increase continuously without reaching any steady-state value, which tends to indicate that debonding and the associated release of the radial residual stresses present from processing are likely to take place up to failure.

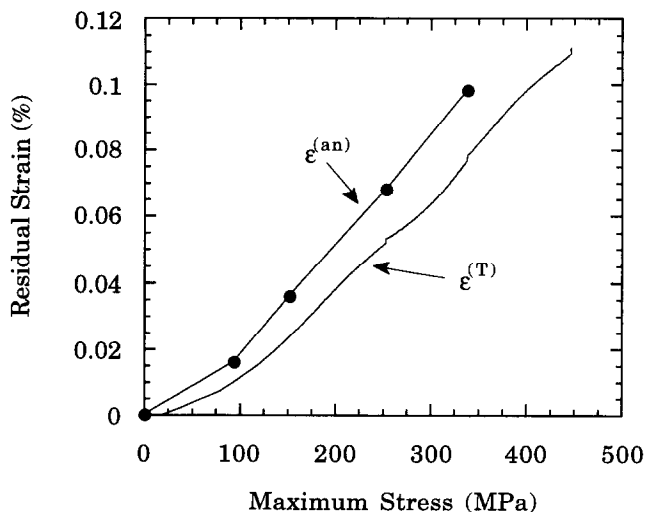
Loading/unloading cycles reveal the consequences of damage, i.e. a progressive decrease of the material's longitudinal Young's modulus along with an extension of residual strains. Residual strains obtained upon unloading in CMCs may be attributed to the interaction of several phenomena, which are depicted schematically in Fig. 5: (i) the release of axial residual stresses (i.e. parallel to the loading direction) present in the composite from processing;<sup>20,21</sup> (ii) partial irreversible sliding arising from the various energy-dissipative frictional mechanisms (and illustrated by a hysteresis loop);<sup>18</sup> and (iii) a mechanical impediment of complete crack closure possibly related to fiber roughness and/or the contact between microcrack lips slightly removed from their initial positions, resulting in increased stiffnesses at the lower end of the loops.<sup>18</sup> In the present case of the 2D C/SiC composite, weakly extended hysteresis loops account for negligible frictional sliding (as expected from a

**Fig. 4.** Room-temperature tensile stress/strain curves of the 2D C/SiC composite.



**Fig. 5.** Schematic representation of a loading/unloading cycle induced hysteresis loop with associated equivalent modulus and anelastic residual strains  $\epsilon^{(an)}$  in a CMC system (based on both experimental observations<sup>18,19</sup> and theoretical predictions<sup>20</sup>).  $\epsilon^{(T)}$  and  $\epsilon^{(f)}$  represent the permanent strains due to release of the thermal residual stresses and to partial irreversible sliding, respectively.

composite system with an interface under residual tension) whereas crack closure impediment effects are nearly absent. Consequently, and as illustrated by Fig. 6, the relative importance of the values taken by these residual strains appears to be essentially related to the release of the axial tensile residual stresses present in the composite from processing. Values of  $\epsilon^{(T)}$  displayed in Fig. 6 were established by extrapolation of the compliance slopes derived from the loading/unloading cycles and representing the equivalent modulus of the material damaged under monotonic tension (see Fig. 5). These compliance



**Fig. 6.** Development of the anelastic residual strains  $\epsilon^{(an)}$  obtained upon unloading, along with the part taken in these strains by release of the thermal residual stresses  $\epsilon^{(T)}$ , as a function of the applied stress.

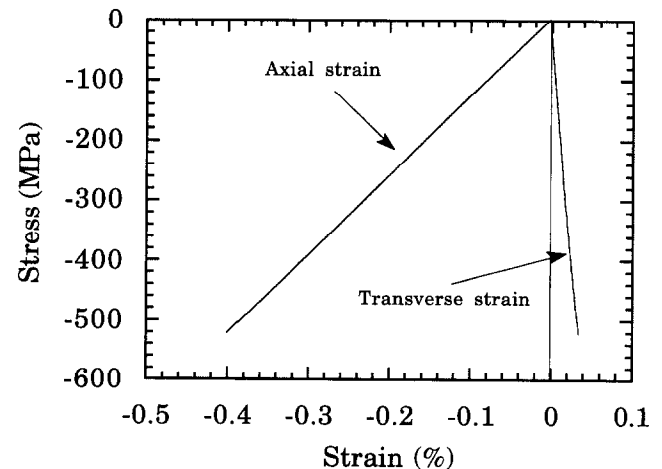
slopes happen to meet at a single point localized in the compression domain, at approximately  $-140 \pm 5$  MPa and  $-0.14 \pm 0.02\%$  (see the dotted lines in Fig. 9). Thus, the monotonic tensile behavior of the C/SiC composite may be considered as damageable-elastic with respect to a fictitious thermal-residual-stress-free origin of the stress/strain axis.

### 3.2.2 Compressive behavior and tension/compression test

A typical example of a compressive stress/strain curve is shown in Fig. 7. In contrast to the tensile behavior, after a slightly non-linear initial portion involving the closing of the transverse thermal microcracks, the curve remained linear almost up to the ultimate stress level. This behavior indicates that (i) once closed and as far as damage is concerned, existing cracks become passive and (ii) no new cracks were generated except possibly close to the ultimate stress level where the AE signal started to increase significantly. This late onset of damage is probably related to the growth and coalescence of the longitudinal thermal microcracks, leading to a failure mode involving delamination and buckling of the fiber bundles (Fig. 8).

In addition, the compression modulus of 140 GPa derived from the linear part of the stress/strain curve, i.e. when the transverse thermal microcracks are closed, may therefore be considered as a good approximation of the longitudinal Young's modulus  $E_3^0$  of the (fictitious) undamaged material. Indeed, if the exact value of  $E_3^0$  should be obtained with the longitudinal thermal microcracks also being closed (e.g. through a biaxial compression test), their effect, once opened, on the longitudinal Young's modulus is known to be weakly significant.<sup>22</sup>

It is worthy of note that, in spite of using tensile specimens (i.e. specimens not specifically designed for compression tests), no excessive bending could be



**Fig. 7.** Room-temperature compressive stress/strain curves of the 2D C/SiC composite.



Fig. 8. Optical view of a 2D C/SiC specimen fractured in compression.

observed, thanks to good alignment and the high stiffness of the loading device. In fact, compressive stress and strain to rupture values obtained in the present work, i.e.  $-520$  MPa and  $-0.4\%$  respectively, compare well to and even fall in a higher range than values obtained from tests performed with a specific compression testing device on a similar CVI-processed 2D C/SiC composite.<sup>11</sup>

Results of the tension/compression test are plotted in Fig. 9. As expected from the tensile tests, the anelastic residual strains appear to be recovered upon compression, which confirms the accuracy of the hypothesis previously made of an overall damageable-elastic behavior. Conversely and most likely because debonds and longitudinal cracks cannot be closed in compression, transverse strains remain truly irreversible. For similar reasons and contrary to the general mechanical behavior of CMCs depicted in Fig. 5, the thermal-residual-stress-free origin of the stress/strain

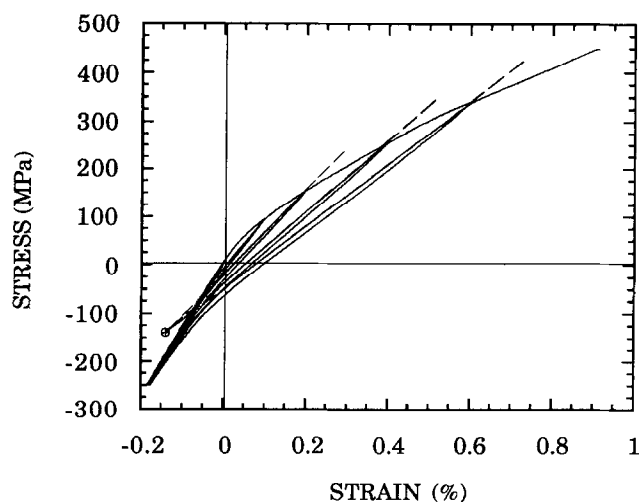


Fig. 9. Room-temperature tensile stress/strain curve with interspersed tension/compression loading/unloading cycles.

axis (i.e. the concurrent point of the compliance slopes) lies out of the stress/strain curve.

### 3.3 Engineering moduli changes induced by damage

Continuous variations of the compliance tensor coefficients  $S_{33}$  and  $S_{32}$  were calculated from the secant slopes joining the thermal-residual-stress-free origins to the longitudinal and transverse strain versus stress curves, respectively. Relative variations of the longitudinal Young's modulus  $E_3$  and the in-plane Poisson's ratio  $\nu_{32}$  then derived from these values (i.e.  $E_3 = 1/S_{33}$ ,  $\nu_{32} = -S_{32}/S_{33}$ ) are plotted in Fig. 10 as a function of the applied stress. The relative variation of the longitudinal Young's modulus is better known as the damage parameter  $D$ , first introduced by Kachanov<sup>23</sup> and generalized in the framework of irreversible thermodynamics by Lemaître and Chaboche.<sup>24</sup> The relative variation of the Poisson's ratio, on the other hand, should not be considered as a damage parameter from the strict point of view of damage mechanics.<sup>15</sup> Nevertheless, useful qualitative information concerning the transverse behavior may be derived. The continuum damage mechanics approach to the mechanical behavior of CMCs and its generalization to the three-dimensional case will be described in detail in Part II of the present study.<sup>13</sup>

The following trends displayed by the relative variation of the longitudinal Young's modulus (or the damage parameter  $D$ ) as a function of the applied stress may be emphasized:

1. as a consequence of the definition retained for the initial modulus  $E_3^0$  of the undamaged material, damage starts to take place in compression. The closing/opening effect of the

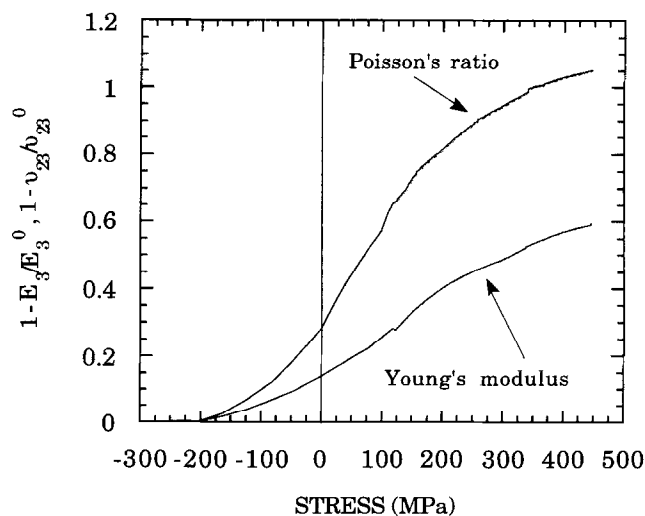


Fig. 10. Relative changes of the axial Young's modulus  $E_3$  and in-plane Poisson's ratio  $\nu_{23}$  as a function of the applied stress.  $E_3^0$  and  $\nu_{23}^0$  are the compression modulus and Poisson's ratio, respectively.

thermal microcracks already present in the material on the Young's modulus is therefore accurately represented, i.e. cracks are 'active' when opened or 'passive' when closed.<sup>25</sup>

2. the Young's modulus then decreases continuously up to final failure of the material but in a proportion less significant than for a 2D woven SiC/SiC composite similarly CVI-processed and failing at a comparable strain of about 0.9% (i.e.  $D \approx 0.6$  and 0.85, respectively<sup>26</sup>). Such a difference tends to underline that the part taken in the various damage mechanisms by transverse microcracking, which is the most effective regarding the Young's modulus decrease, is not as important as in the case of the SiC/SiC composite. This probably has to be related to the difference of fiber/matrix CTE mismatch existing between the two materials, which leads to a state of radial compression at the interface for the SiC/SiC composite and a state of radial tension for the C/SiC composite (i.e. debonding is thus favoured).

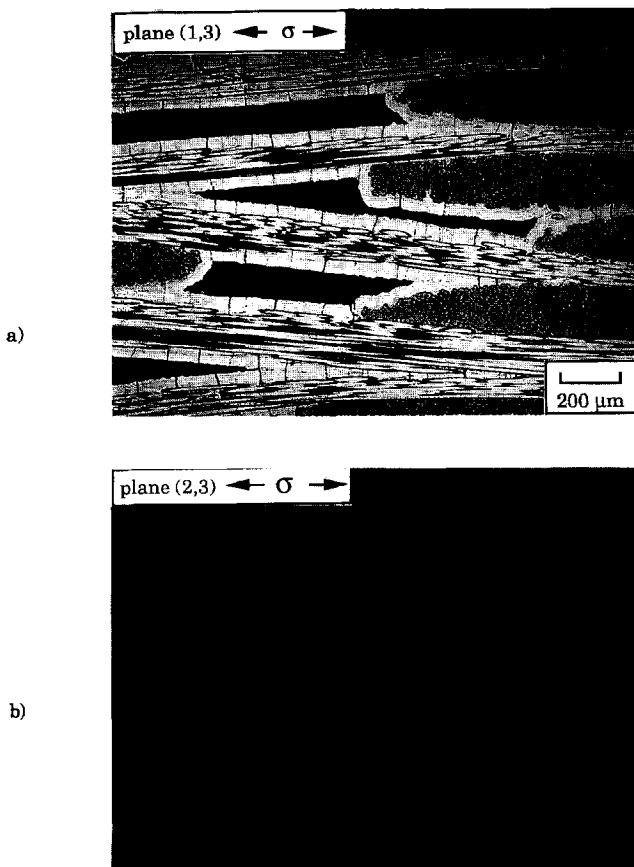
The onset of damage also leads to a continuous

decrease of the Poisson's ratio  $\nu_{32}$ , more important than in the case of the Young's modulus, whereas a negative value is even reached close to rupture. If a zero value for the Poisson's ratio is understandable and accounts for a maximum state of debonding when the matrix does not carry enough load to encounter any Poisson's effect, a negative value is more surprising. This last behavior may be related to a late interaction between the transverse microcracks when debonding is at a maximum and the longitudinal thermal microcracks (which would thus become more opened upon loading than upon unloading).

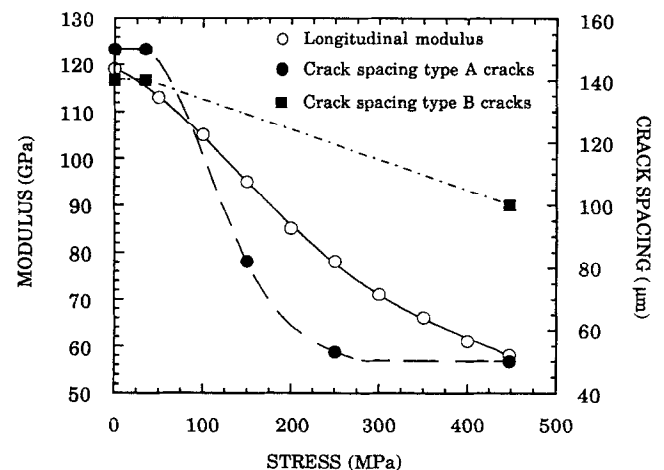
### 3.4 Microstructural evolution of damage and fracture behavior

Figure 11 shows micrographs taken on polished sections of a specimen tested in tension to rupture. The multiplication of transverse microcracks (i.e. perpendicular to the applied load) induced by loading is seen to concern mainly type A cracks whereas the density of type B cracks undergoes only a slight increase (Fig. 11(a)). These few type B cracks newly generated during the tensile test mostly occur at the extremities of the transverse bundles and always coincide with type A cracks from the adjacent longitudinal bundles. Conversely, the density of the other set of thermal microcracks present in the as-received material (i.e. both type A and B cracks parallel to the applied load) proved to be unaffected (Fig. 11 (b)).

The change in crack density as a function of the applied stress, obtained from observations performed on interrupted tensile tests, is plotted in Fig. 12 along with the variation in the longitudinal Young's modulus. Matrix microcracking appears to take place in a relatively narrow stress domain, starting from about 35 MPa (as determined by AE monitoring) and



**Fig. 11.** Micrographs taken on a specimen tested to rupture in tension, showing the multiplication of type A cracks perpendicular to the applied load.



**Fig. 12.** Development of crack spacing (for cracks perpendicular to the applied load) and longitudinal Young's modulus  $E_3$  as a function of the applied stress.

being completed around 200–250 MPa. This result confirms that (i) a large part of damage in terms of transverse matrix microcracks was already present at the end of the processing steps and (ii) the continuous decrease of the Young's modulus (and of the Poisson's ratio, see Fig. 10) which still takes place beyond matrix microcracking saturation and up to composite rupture is representative of the relative importance of other damage mechanisms, such as crack opening and associated sliding and/or debonding and possibly fiber fracture. If frictional sliding is apparently weakly active when considering the limited hysteresis loops obtained upon loading/unloading cycles, debonding, on the other hand, should be largely present. Indeed, microstructural observations performed in both planes (1,3) and (1,2) on specimens fractured in tension also revealed the presence of a significant amount of matrix/bundle and inter-bundle debonds (Fig. 13(a)). These inter-bundle debonds induced longitudinal delamination cracks which were largely present near the fracture surface. Moreover, and most likely as a consequence of the relief of thermal stresses through the intersection of debonding from the transverse cracks with the longitudinal thermal microcracks of

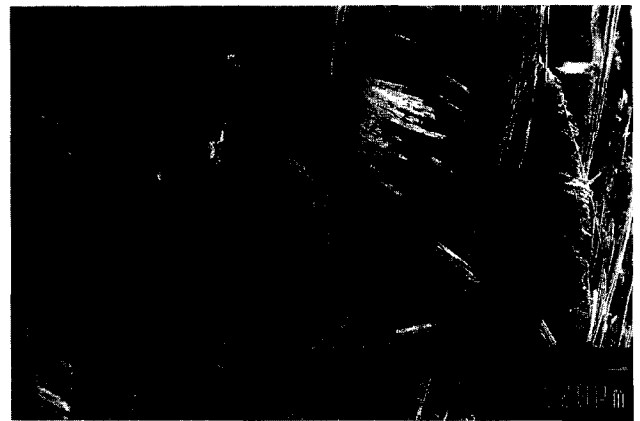


Fig. 14. SEM micrograph showing the fracture surface of a 2D C/SiC specimen tested under tension.

both types A and B, these longitudinal thermal microcracks appeared to be largely opened (Fig. 13(b)).

*In situ* observations revealed breakage of individual fibers starting from an applied load of 350 MPa and a corresponding strain of 0.7%. However, care should be taken with a quantitative interpretation of these values since the fibers observed were lying at the surface of the tensile specimens and had thus been noticeably weakened during polishing.

The fracture surface exhibited in Fig. 14 displays extensive pull-out of the fiber bundles and, within each bundle, of the carbon fibers. In this latter case, however, a closer look reveals that the apparent pull-out length of the external fibers is more related to the disappearance of the surrounding matrix during the impact produced by final rupture than to intra-bundle debonding mechanisms. Bundle pull-out, on the other hand, is likely the direct consequence of the importance of the matrix/bundle and inter-bundle debonding phenomena taking place during the tensile test.

#### 4 SUMMARY AND CONCLUSION

The mechanical behavior of a CVI-processed 2D woven C/SiC composite has been proved to be strongly influenced not only by the onset and propagation of damage induced by the mechanical loading, like most CMCs, but also by the array of thermal microcracks and the thermal residual stresses both present from processing and related to the important CTE mismatch existing between the constituents of the material.

In this respect, the stress/strain response of specimens loaded in tension parallel to fibers is quasi non-linear from the beginning and suggests, with the help of the underlying microstructural changes, three or four stages in the development of damage: (i) an elastic response of the composite coupled with partial

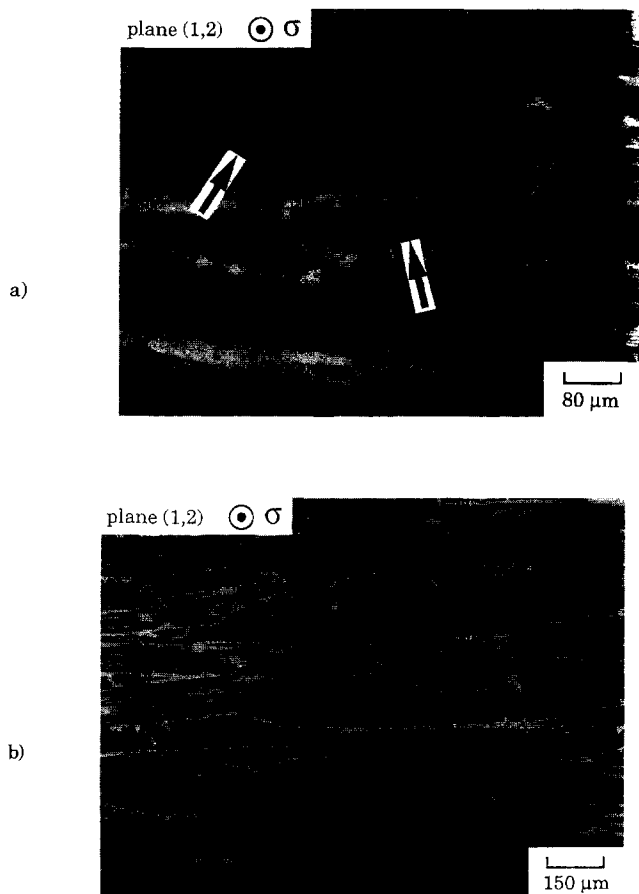


Fig. 13. Micrographs taken on a specimen tested to rupture in tension, showing: (a) longitudinal (delamination) cracks, (b) largely opened longitudinal thermal microcracks.



re-opening of thermal microcracks up to an applied stress of about 30–40 MPa; (ii) multiple matrix microcracking normal to the applied load up to about 200 MPa; (iii) crack opening and related fiber/matrix and (mostly) bundle/matrix and inter-bundle debonding with (iv) progressive load transfer to the fibers which then fracture increasingly until composite failure, at about 455 MPa.

The important residual strains which are observed upon unloading are essentially related to release of the thermal residual stresses and the tensile behavior may thus be considered as damageable-elastic with respect to a fictitious thermal-stress-free origin of the stress/strain axis lying in the compression domain. In compression, after an initial (slightly) non-linear response involving the closure of thermal microcracks, the composite then displays a linear-elastic behavior until failure.

In addition to the development of the transverse crack density which does not account for all the damage mechanisms, the onset and propagation of damage in the material were measured at the macroscopic scale by the induced changes of the Young's modulus and the Poisson's ratio, which both display a continuous decrease until rupture. However, because the so-called transverse microcracks have in fact overriding propagation directions and are accompanied by longitudinal discontinuities such as debonding and delamination, their effect should be highly anisotropic and might affect the whole compliance tensor. Such a behavior, which would have important design implications, will be addressed in the second part of this study by means of an ultrasonic method.

## ACKNOWLEDGEMENTS

The authors are indebted to the SEP for providing the C/SiC material and to B. Humez for performing the mechanical tests.

## REFERENCES

1. Choury, J. J., Thermostructural composite materials: Fabrication and main applications. In *Proc. 4th Int. Symp. on Ceramic Materials and Components for Engines*, ed. R. Carlsson. Chapman and Hall, London, 1992, pp. 102–112.
2. Heraud, L. & Spriet, P., High toughness C–SiC and SiC–SiC composites in heat engines. In *Whisker and Fiber Toughened Ceramics*, ed. R. A. Bradley, D. E. Clark, D. C. Larsen & J. O. Stiegler. ASM International, Metals Park, OH, 1988, pp. 217–224.
3. Courtois, C., Desmaison, J. & Tawil, H., Protection against oxidation of C/SiC composites with a multilayer CVD coating (SiC/TiB<sub>2</sub>/Si<sub>3</sub>N<sub>4</sub>). In *Proc. 3rd Euro-Ceramics Conf.*, ed. P. Duran & J. F. Fernandez. Faenza Editrice Iberica, Castellon de la Plana, 1993, pp. 803–808.
4. Goujard, S., Vandenbulcke, L. & Tawil, H., The oxidation behaviour of two- and three-dimensional C/SiC thermostructural materials protected by chemical-vapour-deposition polylayers coatings. *J. Mater. Sci.*, **29** (1994) 6212–6220.
5. Jouin, J. M., Mechanical behavior under cyclic loading of bidirectional silicon carbide matrix composites. In *Matériaux Composites pour Applications à Hautes Températures*, ed. R. Naslain, J. Lamalle & J. L. Zulfian. AMAC/CODEMAC, Bordeaux, 1990, pp. 168–180 (in French).
6. Lamouroux, F. & Camus, G., Oxidation effects on the mechanical properties of 2D woven C/SiC composites. *J. Eur. Ceram. Soc.*, **14** (1994) 177–188.
7. Evans, A. G. & Marshall, D. B., The mechanical behavior of ceramic matrix composites. *Acta Metall.*, **37** (1989) 2567–2583.
8. Quenisset, J. M., Damage mechanisms in ceramic matrix composites reinforced by continuous fibers. In *Advanced Structural Materials Vol. 2, Proc. 2nd European Conf. on Advanced Materials and Processes*, eds T. W. Clyne & P. J. Withers. Institute of Materials, London, Brookfield, VT, 1991, pp. 418–428.
9. Lamouroux, F., Camus, G. & Thebault, J., Kinetics and mechanisms of oxidation of 2D woven C/SiC composites: I. Experimental approach. *J. Am. Ceram. Soc.*, **77** (1994) 2049–2057.
10. Gomina, M., Fourvel, P., Osterstock, F. & Chermant, J. L., Crack propagation resistance of a fibrous C/SiC composite (in French). *Rev. Phys. Appl.*, **23** (1988) 183–191.
11. Wang, Z. G., Laird, C., Hashin, Z., Rosen, B. W. & Yen, C.-F., Mechanical behaviour of a cross-weave ceramic matrix composite. *J. Mater. Sci.*, **26** (1991) 4751–4758.
12. Shuler, S. F., Holmes, J. W., Wu, X. & Roach, D., Influence of loading frequency on the room-temperature fatigue of a carbon-fiber/SiC-matrix composite. *J. Am. Ceram. Soc.*, **76** (1993) 2327–2336.
13. El Bouazzaoui, R., Baste, S. & Camus, G., Development of damage in a 2D woven C/SiC composite under mechanical loading: II. Ultrasonic characterization. *Comp. Sci. Technol.*, **56** (1996) 1373–1382.
14. Thebault, J., Fabrication process of a composite material with a refractory reinforcement and a ceramic matrix and structure elaborated with such process. French patent 2 567 874 A1, 20 July 1984.
15. Baste, S. & Gerard, A., Evolution of Poisson's ratio and anisotropic damage. In *Mechanics and Mechanisms of Damage in Composites and Multimaterials, Mecamat 89*, ed. D. Baptiste. Mechanical Engineering Publications, London, 1991, pp. 293–305.
16. Wang, S. W. & Parvizi-Majidi, A., Experimental characterization of the tensile behaviour of Nicalon fibre-reinforced calcium aluminosilicate composites. *J. Mater. Sci.*, **27** (1992) 5483–5496.
17. Kim, R. Y. & Pagano, N. J., Crack initiation in unidirectional brittle-matrix composites. *J. Am. Ceram. Soc.*, **74** (1991) 1082–1090.
18. Bouquet, M., Birbis, J. M. & Quenisset, J. M., Toughness assessment of ceramic matrix composites. *Comp. Sci. Technol.*, **37** (1990) 223–248.
19. Thomas, W. A., Kopp, M. W. & Sanchez, J. M., Stress-strain behavior of Nicalon-fiber-reinforced calcium aluminosilicate composites under tensile fatigue conditions. *J. Am. Ceram. Soc.*, **76** (1993) 2175–2179.

20. Gudmundson, P. & Ostlund, S., Prediction of thermoelastic properties of composite laminates with matrix cracks. *Comp. Sci. Technol.*, **44** (1992) 95–105.
21. Kuo, W. S. & Chou, T.-W., Modeling of the thermomechanical behavior of glass matrix composites. In *Metal and Ceramic Matrix Composites: Processing, Modeling and Mechanical Behavior*, ed. R. B. Bhagat et al. The Minerals, Metals and Materials Society, Warrendale, PA, 1990, pp. 311–318.
22. Krajcinovic, D., Continuum damage mechanics. *Appl. Mech. Rev.*, **37** (1984) 1–6.
23. Kachanov, L. M. On the creep rupture time. *Izv. AN SSSR, Otd. Tekh. Nauk*, **8** (1958) 26–31.
24. Lemaître, J. & Chaboche, J. L., Phenomenological aspect of damage induced fracture (in French). *J. Mec. Appl.*, **2** (1978) 167–189.
25. Baste, S., El Guerjouma, R. & Audoin, B., Effect of microcracking on the macroscopic behavior of ceramic matrix composites: Ultrasonic evaluation of anisotropic damage. *Mech. Mater.*, **14** (1992) 15–31.
26. Guillaumat, L., Microcracking of CMCs: Relationships with the microstructure and the mechanical behavior. PhD thesis 1056, University of Bordeaux, Bordeaux, France, 1994.